Problem Chosen	2026 HiMCM Summary Sheet	Team Control Number
A		18213

Choosing a City for High-School Life with Incomplete Evidence

Summary

A city recommendation becomes more useful when its evidence and priorities can be examined together. We compare sixteen cities using ten indicators of safety, environment, household costs, education and everyday resources. Zurich is the conservative recommendation under the retained team inputs and baseline or equal weights. All sixteen cities remain possible champions within the retained evidence bounds.

The evaluation combines fixed-anchor TOPSIS with a maximum-regret decision. Benefit and cost utilities share fixed reference scales, allowing controlled comparisons across data, preference and development scenarios. Missing values remain intervals. We select the city with the smallest worst-case shortfall from a better-scoring competitor over the declared rectangular uncertainty set.

Official municipal records, school statistics and operator inventories supplement twenty-three previously missing entries while preserving the team's original values. The average score-interval width falls from 18.81077 to 16.04938, a 14.68% reduction. Actual reference years and source populations remain explicit.

The interval revision retains 48 uncertain entries and Zurich's recommendation, while 13 fixed-weight ranks change from midpoint completion. Separate interior completions let each city rank first. Withholding Zurich's homicide point selects Singapore under baseline weights and Oslo under equal weights. Separately withdrawing its work-commute proxy selects Singapore under baseline weights; equal weights retain Zurich. These tests show dependence on accepted inputs.

Preferences and development conditions answer different decision questions. Doubling the relative emphasis on education changes the conservative choice to Oslo. In 10,000 baseline weight perturbations at Dirichlet concentration 100, Zurich is recommended in 43.33% of draws, compared with 27.92% for Oslo and 28.23% for Singapore. These are conditional simulation frequencies. Three uniform ten-per-cent development scenarios retain Zurich; separate city-specific shocks reveal conditional recommendation changes without claiming an empirical forecast.

Separate country, district and age-group cases examine what citywide totals leave unresolved. They distinguish facility supply from student access and observed broad age groups from unsupported age-specific preferences. The result is a practical starting choice with explicit alternatives and clear priorities for additional evidence.

Keywords: high-school opportunities; TOPSIS; interval evidence; maximum regret; preference sensitivity.

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1 Introduction and decision scope

1.1 What makes a city suitable for high-school life?

A useful comparison connects a student's school day with the opportunities and constraints around it. Safety and air quality affect the daily setting; teaching resources and nearby higher education describe educational supply; housing, living costs, transport, travel, broadband and swimming facilities describe household pressures and everyday resources. Our operational definition of "best" is the conservative choice under these ten criteria, declared weights and retained evidence ranges. It provides a starting recommendation for the compared cities, rather than a claim about every student's experience.

The study uses the sixteen cities in the frozen team matrix: Amsterdam, Berlin, Dublin, Helsinki, London, Madrid, Milan, Oslo, Paris, Prague, Singapore, Stockholm, Tokyo, Vienna, Warsaw and Zurich. This is an inherited comparison set across sixteen countries, not a representative sample of all world cities. It exceeds the problem's minimum of ten cities and spans different resource and cost settings. The findings concern this set; a broader claim would require an explicit sampling design and additional comparable observations.

1.2 Criterion selection and contribution

The ten indicators preserve the agreed team framework and balance education with the conditions that make resources usable in daily life. Their directions and anchors appear in Table 1. Counts and city averages serve as scoped supply or setting measures. They do not directly measure friendships, student wellbeing, school quality, language fit, discrimination, cultural participation or individual access. Those outcomes are relevant to a family's choice; comparable student-level observations are needed to integrate them without inventing measurements or double-counting existing proxies.

Our central contribution is a city decision that retains incomplete evidence. We improve source precision, compare a point-completion benchmark with an interval revision, and examine preferences and future conditions without changing the primary indicator set. TOPSIS provides one main quality score; AHP structures stated priorities, Monte Carlo measures specified perturbations, and VIKOR supplies a focused method comparison. The extensions test the recommendation rather than vote among an expanding set of ranking models.

AI assistance supported research retrieval, calculations, code and prose preparation, including Exa source retrieval and extraction [18, 19]. The appended use report records the known assistance and the outstanding documentation required before submission.

2 Assumptions and their role

1. **Comparability follows the retained scope.** Original values and population denominators remain fixed. Each supplement keeps its provider, year, city boundary and educational population. Mixed-year densities are conditional comparisons; they do not create a same-year census.

2. **Utilities are monotone on fixed scales.** More resources and less cost or burden improve their respective utilities. Declared anchors set the scale; they are neither estimated preferences nor empirical optimum values.

3. **The score permits compensation.** Weighted distances combine criteria in a common utility space. The model therefore allows stronger results on some criteria to offset weaker results on others. A family requiring a hard safety or eligibility threshold should apply it separately before choosing.

4. **The evidence set is rectangular.** Uncertain utilities vary independently within retained intervals. This conservative closure omits unknown cross-criterion and cross-city restrictions. Its possible champions and regret bounds are conditional on this set; they are not population probabilities.

5. **Extensions hold the comparison basis fixed.** Scenario changes retain population denominators, anchors and the indicated weight rule. They are conditional experiments with no fitted future date or representative age-specific preference distribution.

3 Notation

Symbol	Meaning
M, N	Sixteen cities and ten indicators
w_j, a_j	Normalized criterion weight and fixed utility anchor
u_{ij}	Utility of indicator j for city i, on $[0, 1]$
ℓ_{ij}, h_{ij}	Retained lower and upper utility bounds
L_i, U_i	Lower and upper TOPSIS score, on $[0, 100]$
$R_i, \mathcal{C}$	Maximum regret and set of possible champions
κ	Concentration of a declared Dirichlet weight distribution
r	Raw-value multiplier in a development scenario

4 Data collection and comparability

4.1 A traceable city–indicator matrix

We compare $M = 16$ cities on $N = 10$ indicators covering safety, environment, household costs, education and everyday resources. The initial matrix is the team-supplied city table [1]. We retain its existing values, populations, reference years and source flags as the agreed input, then supplement only empty cells. Some supplied values lack original source URLs, so their provenance remains the team table; official supplements do not retrospectively validate every supplied measurement. Table 1 states what each variable measures.

The nominal reference year is 2025; supplements retain their actual years and supplied population denominators. Matching Eurostat records give Dublin's 28.8 minutes for 2011 and Zurich's 39.9 for 2024 as journeys to work, currently labelled "greater city"; Zurich's

Table 1: Retained indicators, units and reference anchors. Supplied values are retained; supplements have explicit source scope.

Indicator	Measurement and interpretation	Anchor
Homicide	Deaths per 100,000; cost; supplied city proxy	2
$PM_{2.5}$	$\mu g/m^3$; cost; supplied concentration proxy	15
Living costs	Supplied living-cost index; cost	100
Housing	Supplied price-to-income ratio; cost	10
School resources	Pupils per teacher or explicitly scoped FTE; cost	15
Higher education	Institutions per 100,000; benefit; supply proxy	1
Transport stops	Unique stops per 100,000; benefit; partial inventories	50
Travel	Duration in minutes; $\exp(-t/30)$; commute or stage proxy	30 min
Broadband	Supplied Mbps; benefit; city/general-use proxy	100
Swimming pools	Venues per 100,000; benefit; supply proxy	1

value is flagged estimated [53]. These matches establish neither the team's source lineage nor the historical boundary. We retain the supplied means, years and flags as conditional general-commute proxies. They do not identify high-school journeys. Mixed-year densities and historical inventories remain conditional; withdrawal scenarios examine their role in the decision.

We use official statistical tables, operator inventories and municipal geodata to fill missing fields. For each admitted supplement, the archive records the provider, URL, year, geographic and population scope, units, extraction rule and file hash. Original PDFs and API responses are retained separately from derived tables. Search summaries locate evidence; numerical admission requires the underlying record or an explicitly identified provider capture. Values are converted to compatible units before utility calculation: for example, the supplied higher-education density per million is divided by ten to obtain the per-100,000 scale. Mode share is not converted into physical stop density.

4.2 Collection problems and their resolution

Count city facilities, not platforms or entire networks. Selecting Oslo's municipal GIS records within the city boundary yields 89 metro stations after consolidating 186 platforms; Zurich's 2025 VBZ report distinguishes 446 city stops from 235 outside the city [5, 6]. Helsinki's November 2026 operator snapshot similarly separates 19 Helsinki stations from 11 in Espoo [7]. For a verified subset count c, fixed population P and anchor a, we use $d = 10^5 c/P$ and utility $[d/(a+d), 1]$, leaving uncounted providers unknown. Madrid's May 2026 register has 293 records, of which 242 carry city code 28079. Omitting two unnamed records without lines and consolidating native station identifiers gives 191 complexes [23, 24]. Vienna illustrates why record totals alone are insufficient: 1,695 named areas include unverified depot objects, so only 99 metro areas individually matched to existing-station geodata and the city polygon are retained [28, 29].

Native identifiers, verified aliases and operator interchange maps prevent transfer stations from being counted once per line. Prague retains 57 current metro complexes; Flora's closure explains its omission from the 58 distinct complexes in the 2025 report, whereas Florenc's partial accessibility restriction does not close the whole complex [30, 31, 32]. Berlin's nine BVG U-Bahn profiles consolidate 131 zone-A appearances into 100 complexes. Zone A extends through the S-Bahn ring; the retained U-Bahn subset leaves outer stations and other modes uncounted [33]. Milan's municipal layer, distinguished from its entrance layer, selects records with both the Milano name and code 015146, yielding 104 stations/hubs after five map-supported interchange consolidations; proximity alone does not merge stations [47]. Amsterdam's municipal map contains 39 Metro-designated station groups. Thirty-five lie inside both the municipal 2026 and CBS 2025 city boundaries; a declared 100-metre nominal boundary screen omits three and retains 32 [51, 52]. The screen defines an analyst-selected subset, not a provider positional-error guarantee. Nearby boarding points are already grouped by the provider. The inventory metadata specify 2026 but no exact observation date; pairing it with the retained 2025 population of 941,927 gives a conditional lower bound, with other transport supply left unknown.

London's inventory mixes platforms and entrances with stations. From 272 station-level records, 62 have exact fare-zone value 1; it retains 59 stations/hubs after two conservative omissions and Paddington hub consolidation; representative coordinates lie within the ONS December 2025 London-region boundary [36, 37]. Singapore's LTA GeoJSON contributes 5,205 distinct nonempty bus-stop codes, without distance clustering [41]; Stockholm's 2025 city–police report identifies thirteen metro stations in two districts [46].

Keep inventory dates, status and venue units explicit. Milan's layer is dated 16 October 2024; Paris's 244 intra-muros metro stations come from a March 2022 atlas using RATP 2021 and Apur 2022 sources [47, 9]. These historical lower-bound proxies are not relabelled as 2025 counts; a separate withdrawal restores the Paris supplement to unknown. Vienna, Berlin and London use November 2026 inventories. All retain the supplied population denominators, including Paris's 2023, Dublin's 2022 and Singapore's 2024 values. Inventory membership does not establish continuous operation or adolescent access. Stockholm's November 2026 directory retains thirteen municipal swimming venues after excluding a renovation closure and counting its replacement once [8]. Tokyo retains five Nerima indoor venues at distinct addresses: Heiwadai's current closure excludes it, while Miharadai's announced February–May 2027 closure is not a current closure [25, 26]. Other named pool subsets contain seven Singapore venues from April 2025, seven Zurich indoor venues accessed in November 2026, and seventeen Milan indoor venues under the June 2025 tariff [42, 44, 45]. They count physical venues, not basins or lanes, and leave other providers unknown.

Match school stages and teaching units before forming ratios. Prague's 2024/25 totals are 80,073 secondary-school pupils and 6,877.1 teacher full-time equivalents (FTE), including lower-gymnasium years. Subtracting the matching 8,748 pupils and 737.8 FTE gives 71,325 upper-stage pupils and a rounding envelope of $[6139.2, 6139.4]$ teacher FTE, including adult and non-daytime study modes [10]. Stockholm's city-school ratio cov-

ers upper-secondary years 1–3, all ownership types and the 15 October reference date. Its API value 16.42091405 is retained as $[16.35, 16.45]$: a declared precision convention, not a measurement-error confidence interval [11]. Warsaw's 2024/25 yearbook gives 119,740 pupils across six youth school routes by school location and all ownership types. Unknown youth allocation within adult-inclusive general-school FTE gives 2,490–8,144 FTE; outward one-FTE guards on four published cells widen this to 2,487–8,148 [43]. The guards encode a unit-rounding assumption. These allocation and rounding bounds describe school resources, not a resident age-14–18 population or a statistical confidence interval.

Some published denominators fail these checks. Henri-IV in Paris reports 794 lycée pupils and 153.65 institution FTE for 2025, without upper-stage allocation or complete city coverage; a 2026 post-baccalaureate directory flag cannot recover 2025 workloads [34, 35]. Dublin's school-roll payroll FTE are unpublished [27]; its November 2025 City Of Dublin ETB post-primary vocational subgroup has 411.8 payroll FTE against 376.1 allocated posts, so allocations are not a payroll ceiling [48]. Finland's national FTE combines register, salary and private-provider estimation; without a verified one-sided error envelope it is not a guaranteed Helsinki teacher upper bound [49]. All three city PTRs remain unknown.

Preserve incompatible rate units and supplied values. Amsterdam's CBS sources give ten provisional victims in 2022 and 68 across 2018–2022, neither matching the supplied numerator 340 [39]. Zurich's PKS lists one completed offence in 2024 and two in 2025; offences count criminal acts independently of victims, and the submitted rate 0.14 still lacks a verified denominator [40]. Both original entries remain, with their contribution tested in Section 5.5; we do not manufacture replacement rates from incompatible units.

Retain unobserved travellers and failed retrievals as uncertainty. Singapore's initial census-PDF endpoint returned 404; the official Parliament mirror and API supplied the matching report and table [13]. Duration-bin utility uses endpoints, no midpoints or artificial open-tail cap. Weighted enrolment estimates exceed travel coverage by 1,811 secondary, 989 post-secondary non-tertiary and 1,635 polytechnic students. These are weighted survey estimates, not complete headcounts. Uncovered mass retains $[0, 1]$ utility, and unknown stage shares use the interval convex hull. The resulting $[0.21048, 0.61884]$ is a 2020 school-stage proxy, not an identified 2025 interval for ages 14–18. Dublin's stop-download 403 yielded no catalogue-inferred count; a later official map from the 2023 plan, issued in 2024, supports nineteen existing city-centre tram stops [14]. Warsaw's recovered 2022 survey reports a rounded 27-minute school trip for a city subgroup with sample base 22, but covers residents aged 15 and above without an upper-secondary or matched 14–18 denominator [38]. Its travel utility remains unknown.

Figure 1 locates the remaining uncertainty. The current 160 cells comprise 112 point values, 35 partial intervals and 13 fully missing intervals: eight teacher-ratio, two transport and three travel entries. The original table had 112 points, 12 partial intervals and 36 fully missing cells. The twenty-three supplements therefore improve information without replacing the team's existing measurements. Facility supply, general commute proxies

and school resources remain distinct from adolescent access, school travel and learning quality.

5 Establishing the baseline evaluation model

5.1 Determining indicator weights

We retain eight main indicators at weight 0.1125 each and two auxiliary facility indicators, transport stops and pools, at 0.05 each, so $\sum_j w_j = 1$. These weights express a transparent modelling choice. They are not estimated from a student survey. Equal weights provide a separate reference. TOPSIS is the primary score; VIKOR provides a focused method comparison. AHP represents explicit preference scenarios [2], and Monte Carlo measures sensitivity to declared distributions. These extensions retain the same criteria rather than introduce unrelated ranking models.

For preference multipliers $m_j > 0$, we use

$$w'_j = \frac{m_j w_j}{\sum_k w_k m_k}, \qquad A_{jk} = \frac{w'_j}{w'_k}. \tag{1}$$

The principal eigenvector of A recovers w'. With $\lambda_{\max}$ its largest eigenvalue, $CI = (\lambda_{\max} - N)/(N-1)$ and $CR = CI/RI_N$. Because this matrix is built from exact ratios, $\lambda_{\max} = N$ and $CR = 0$ up to numerical precision. This verifies internal consistency; it does not establish that these are students' actual preferences.

5.2 TOPSIS scoring calculation

We adapt TOPSIS [3] using fixed reference anchors to preserve a common scoring scale across evidence, preference and future scenarios. For benefit and cost measurements respectively,

$$u^+(x) = \frac{x}{a+x}, \qquad u^-(x) = \frac{a}{a+x}. \tag{2}$$

Each anchor in Table 1 is a declared reference scale, not an estimated optimum. Travel uses $q(t) = e^{-t/30}$ for duration t in minutes. When only mean duration $\bar{t}$ is known, Jensen's inequality gives $\mathbb{E}[q(t)] \in [e^{-\bar{t}/30}, 1]$. Fully missing utilities remain $[0, 1]$.

For utility vector u_i, weighted distances from fixed worst and best points are

$$D_i^- = \sqrt{\sum_j w_j u_{ij}^2}, \qquad D_i^+ = \sqrt{\sum_j w_j (1-u_{ij})^2}, \qquad S_i = 100\frac{D_i^-}{D_i^- + D_i^+}. \tag{3}$$

Weights enter the squared distances once. Since S_i increases with every utility, endpoints ℓ_i, h_i give $L_i = S(\ell_i)$ and $U_i = S(h_i)$. Under the retained rectangular uncertainty set, define

$$R_i = \max\{0, \max_{k \neq i} U_k - L_i\}, \qquad i^* \in \arg\min_i R_i, \tag{4}$$

$$\mathcal{C} = \{i : U_i \geq \max_{k \neq i} L_k\}. \tag{5}$$

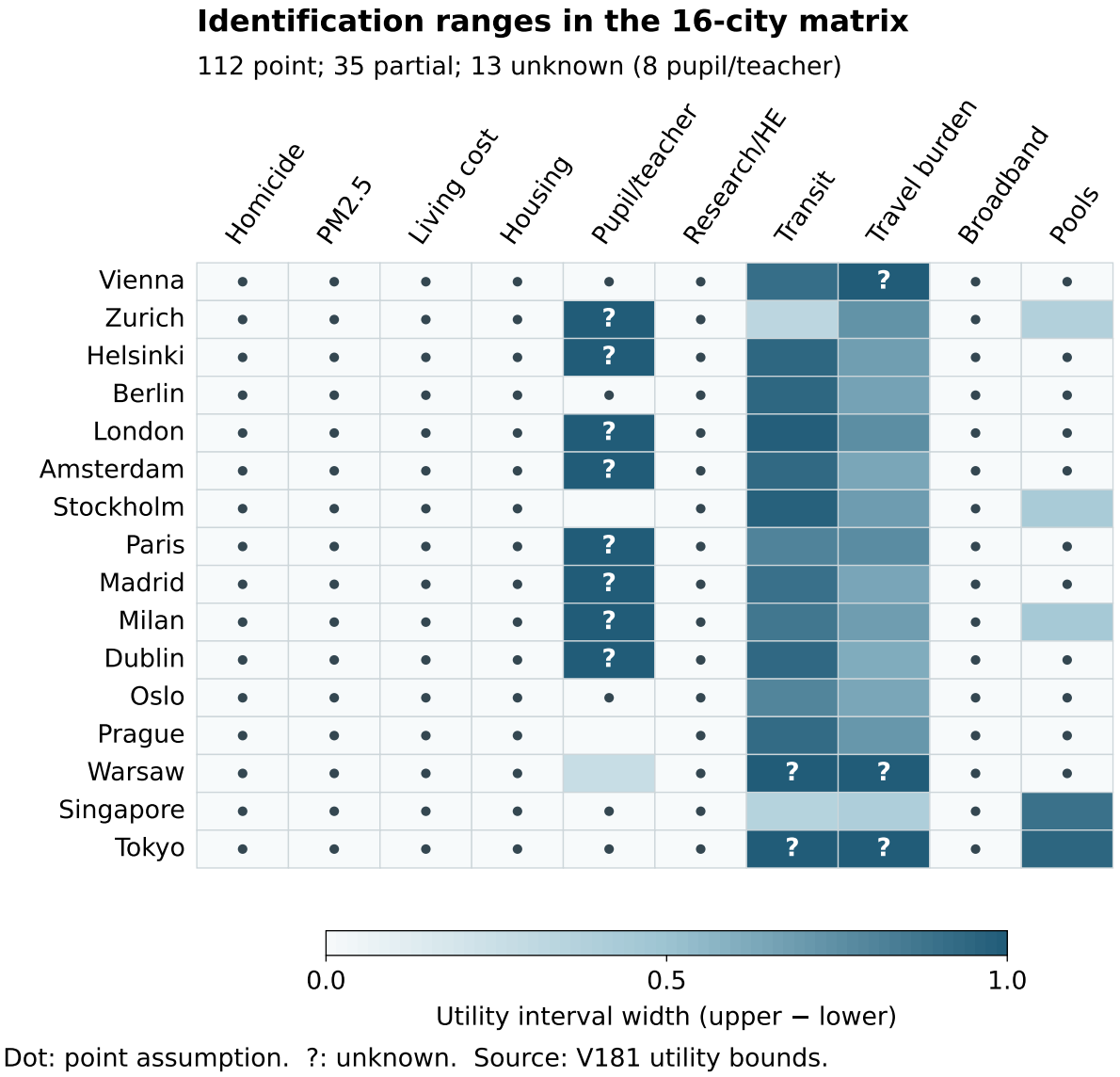


Figure 1: Identification ranges in the retained sixteen-city, ten-indicator matrix. Eight of the thirteen fully unknown entries concern upper-secondary pupil–teacher ratios. Shading shows utility upper minus lower bounds; dots mark retained point assumptions and question marks identify $[0, 1]$ entries. Point assumptions include submitted values and do not certify independent validation or zero measurement error. These are identification bounds under the retained source rules, not confidence intervals. Travel entries retain school-travel and general-commute proxy scopes. Cities follow the input order. Source: V181 utility bounds.

Maximum regret R_i measures how far the selected city could trail another city. Set $\mathcal{C}$ retains every city that could still be first. The maximum excludes the selected city itself. These conditional data bounds are not sampling confidence intervals, and a conservative recommendation does not identify a uniquely best city.

5.3 Scoring results and interpretation

Zurich minimizes maximum regret, followed by Singapore and Oslo (Table 2). Zurich's score lower bound is below those of Singapore and Oslo; its own upper bound is excluded from its competitor set. Its regret is $70.90353 - 53.53548 = 17.36805$, whereas Oslo's is $72.18379 - 54.29594 = 17.88785$. All sixteen cities remain possible champions. The decision order expresses a conservative recommendation under the stated criteria and intervals.

Table 2: Conservative baseline decision ranks with unchanged weights.

Rank	City	Score interval	Maximum regret
1	Zurich	53.54–72.18	17.37
2	Singapore	54.33–63.53	17.85
3	Oslo	54.30–63.90	17.89
4	Stockholm	52.12–64.54	20.06
5	Prague	51.97–62.78	20.21
6	Dublin	51.13–70.90	21.05
7	Berlin	50.73–61.54	21.45
8	Helsinki	49.41–69.84	22.78
9	Paris	49.11–69.04	23.08
10	Madrid	49.00–68.22	23.18
11	Vienna	47.52–61.03	24.66
12	Warsaw	47.50–63.99	24.69
13	Milan	46.07–65.99	26.11
14	London	45.22–66.12	26.97
15	Tokyo	43.96–61.12	28.22
16	Amsterdam	42.48–60.43	29.70

Holding weights and formulas fixed, the twenty-three admitted supplements narrow the average fixed-weight score interval from 18.81077 to 16.04938, a 14.68% reduction. The Amsterdam mapped Metro subset narrows the preceding fixed-weight mean from 16.06144 to 16.04938. The 32 retained station groups raise one utility lower bound while all static decision ranks remain unchanged. The 2026 inventory metadata are paired conditionally with the retained 2025 population; this is an information update, not an observed improvement over time. Earlier additions improved Stockholm's decision position. Withdrawal of the historical Paris and Dublin transport inventories still recommends Zurich; withdrawal of Singapore's historical school-stage travel proxy also preserves this recommendation. Evidence improves precision and changes decisions before it necessarily reduces the possible-champion set.

As a method diagnostic, TOPSIS and VIKOR use the same illustrative midpoint utilities and weights. VIKOR balances total and largest weighted shortfalls equally and uses sample best/worst values [4]. TOPSIS selects Zurich at the midpoint, whereas VIKOR selects Dublin; Kendall's $\tau_b = 0.483$. Dublin satisfies VIKOR's advantage and stability conditions. Under equal weights, $\tau_b = 0.650$; Oslo has the lowest VIKOR index but fails those acceptance conditions, so it is not an accepted unique compromise solution. The midpoint comparison does not replace the interval decision in Equation 4. Agreement indicates similarity within the chosen conventions; disagreement exposes method dependence rather than justifying a post-hoc choice of the preferred winner.

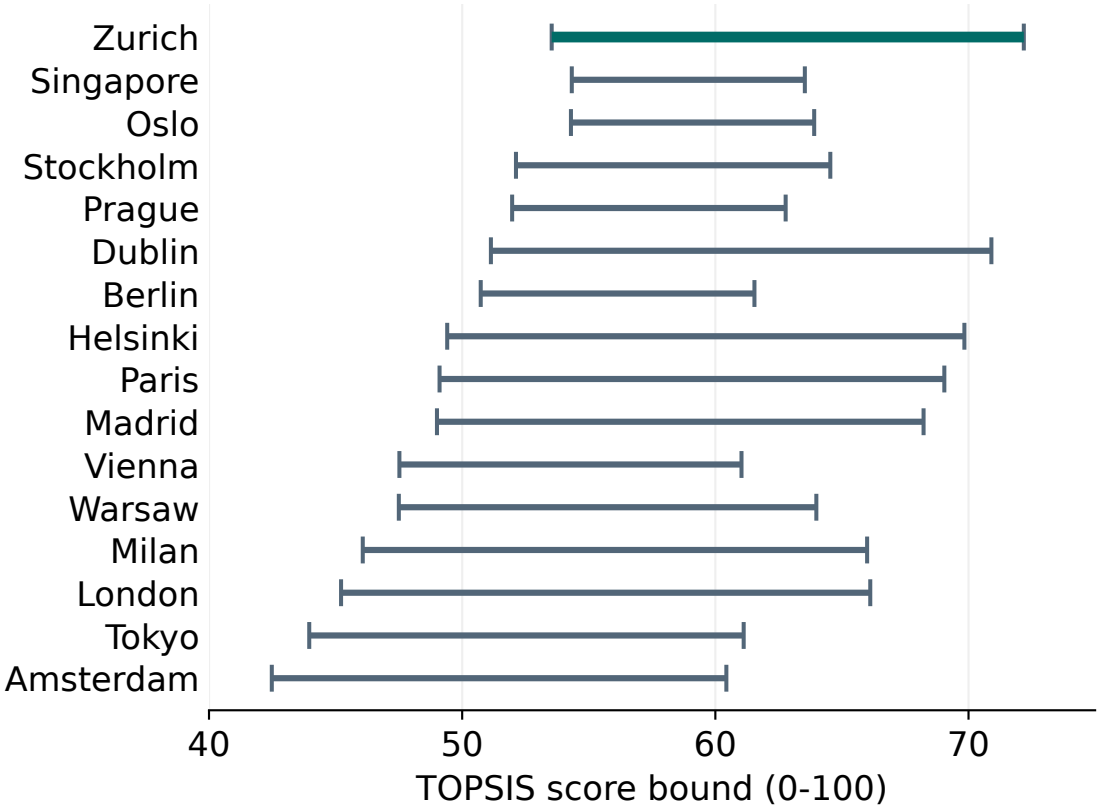


Figure 2: Current score bounds and conservative decision order for the sixteen cities. Intervals retain missing information; they are not confidence intervals. Source: frozen city matrix and admitted supplements.

5.4 A significant revision: retaining uncertain evidence

A controlled comparison shows why evidence bounds belong in the decision. On the same 160 utility entries, compare the midpoint completion $z_{ij} = (\ell_{ij} + h_{ij})/2$ with the retained-interval formulation in Equation 4. Both use identical weights, anchors and TOPSIS scores. The midpoint is a declared diagnostic scenario, rather than an estimate of missing observations or a reconstruction of the team's historical first model.

The revision retains 48 non-point entries and selects the city with the smallest worst-case shortfall. Zurich remains the recommendation, but 13 cities change decision rank under fixed weights; Kendall's $\tau_b = 0.583$. With equal weights, 15 ranks change and $\tau_b = 0.650$.

For each target city c, construct a separate interior completion: set every uncertain utility to $\ell_{ij} + \alpha_i(h_{ij} - \ell_{ij})$, with $\alpha_c = 0.95$ and $\alpha_i = 0.05$ for $i \neq c$. All 112 point values remain fixed and all 48 uncertain values lie strictly inside their intervals. Each target ranks first under both baseline and equal weights. Across the sixteen targets, the smallest lead is 4.974 score points under baseline weights and 5.323 under equal weights. These separate completions establish feasibility in the declared utility rectangle; they carry no estimated probabilities and do not establish joint physical attainability.

Retaining intervals changes the decision order while preserving sixteen possible champions. The 14.68% reduction in interval width comes from admitted data, separately from this formulation comparison.

5.5 Sensitivity to the accepted input information

Point-withdrawal tests keep the primary data, weights, supplements and 48 non-point bounds fixed, replacing selected point utilities by $[0, 1]$. Of 112 points, 64 are dated 2026, two 2025 and 46 earlier. The 28 non-`ok` points comprise 16 `MODELLED`, eight `STALE`, three `SUSPECT` and one `NATIONAL`. Groups cover dates before 2020, after 2025, non-`ok` flags and their union; overlaps count once.

Table 3: Recommendation after withdrawing selected submitted point information. Width is the mean baseline-weight score range on the 0–100 scale.

Withdrawn group	Cells	Baseline	Equal	Width
Retained input	0	Zurich	Zurich	16.05
Before 2020	13	Singapore	Zurich	21.12
After 2025	64	Zurich	Zurich	48.41
Non-ok flags	28	Berlin	Paris	29.11
Union	95	Singapore	Singapore	68.78

Withdrawing 28 flagged points selects Berlin under baseline weights and Paris under equal weights. Removing 13 pre-2020 points selects Singapore under baseline weights and retains Zurich under equal weights. Removing 64 later-dated points retains Zurich under both, widening the baseline mean range from 16.05 to 48.41. The 95-point union selects Singapore under both. All sixteen remain possible champions. Labels and cutoffs are not error probabilities or proof that flagged entries are wrong.

Withholding Zurich's homicide-rate point selects Singapore under baseline weights and Oslo under equal weights; withholding Amsterdam's retains Zurich under both. Other entries remain fixed, without a replacement rate. Varying only Zurich's point rate gives regret ties at about 0.23002 per 100,000 under baseline weights and 0.90473 under equal weights. Below each respective cutoff Zurich uniquely minimizes regret; above it Zurich does not. These are hypothetical cutoffs, not measured rates or confidence limits.

Replace selected commute intervals by $[0, 1]$, holding other entries fixed. Withdrawing Zurich's alone, both identified proxies or all twelve supplied means selects Singapore under baseline weights; equal weights retain Zurich. Dublin alone changes neither choice; all withdrawals retain sixteen possible champions. Separately vary only Zurich's mean with Jensen bounds: baseline weights give a regret tie near 46.59 minutes, selecting Zurich below and Singapore above. Equal weights retain Zurich for all nonnegative means, with other data fixed.

6 Model revision and extension

6.1 Considering future changes

We examine three conditional changes while holding anchors and population denominators fixed. Resource expansion raises higher-education, stop and pool densities by 10%. Living pressure raises living costs, housing cost ratios and travel durations by 10%. Balanced improvement reduces pupil–teacher ratios, $PM_{2.5}$ and travel durations by 10%. A comparable multi-year panel is not available, so these changes have no estimated annual rate or forecast date.

For raw multiplier $r > 0$, the utility transformations are

$$T_r^+(u) = \frac{ru}{1 - u + ru}, \qquad T_r^-(u) = \frac{u}{u + r(1 - u)}. \tag{6}$$

These increasing maps propagate raw-scale changes through benefit and cost intervals. For known mean travel duration, the new Jensen interval is $[e^{-r\bar{t}/30}, 1]$. Singapore instead has bounds $[L_q, U_q]$ on expected utility. With $q = e^{-t/30}$ and $t' = rt$, Jensen and $0 \le q \le 1$ give

$$\mathbb{E}[q^r] \in \begin{cases} [L_q, U_q^r], & 0 < r < 1, \\ [L_q, U_q], & r = 1, \\ [L_q^r, U_q], & r > 1. \end{cases} \tag{7}$$

Raising both endpoints to r would generally misrepresent an expected utility. Equation 7 is an outer moment bound and need not be reversible. Fully missing travel remains unknown. Thus the extension accommodates the different information supplied by means and by travel distributions.

All three uniform-change scenarios retain Zurich as the conservative recommendation and all sixteen cities as possible champions under both weight references, including the Singapore-proxy withdrawal arm. Balanced improvement changes the fixed-weight second and third decisions from Singapore–Oslo to Oslo–Singapore. We also solve a decision boundary: holding other cities fixed, proportionally reduce one city's pupil–teacher ratio, $PM_{2.5}$ and travel durations until its maximum regret ties that of the best other city. Oslo reaches the tie at approximately 5.685% reduction, Singapore at 10.309%, and Stockholm at 28.875%. Zurich already minimizes regret at baseline. A slightly greater reduction produces a strict recommendation switch in these calculations. These thresholds describe the model response to a specified joint change; they do not establish policy feasibility

or a uniquely identified champion. Berlin's hypothetical combined-cost tie threshold is 38.084% after the Amsterdam mapped-inventory update. Scenario definitions and weights are unchanged.

For a separate future-shock simulation, each draw combines a common shock and city-specific shocks under a declared bounded distribution, with the criterion weights held fixed. Original data intervals are not randomly completed. Specifically, $C \sim U[0,1]$ is shared across cities and $I_i \sim U[0,1]$ varies by city, independently; $\delta_i = 0.05C + 0.05I_i$. The relevant raw multipliers are $1 + \delta_i$ or $1 - \delta_i$, and affected criteria within a city share that multiplier. With 10,000 draws per scenario, Zurich's fixed-weight recommendation frequencies are 100.00%, 96.85%, 97.72% for resource expansion, living pressure and balanced improvement respectively. Repeating the same shocks after withdrawing Singapore's historical stage-travel proxy gives 100.00%, 99.19%, 97.72% respectively. These paired calculations isolate dependence on that proxy while retaining the chosen shock distribution. These are conditional stress frequencies under an uncalibrated distribution. Weight perturbations below answer a separate question and are not blended silently into the future-shock frequencies.

6.2 Considering different preferences and development situations

Education priority doubles the weights of the two education indicators. The independent living scenario doubles travel and transport-stop weights. Equation 1 then normalizes all weights. Education priority changes the conservative recommendation from Zurich to Oslo, with decision-rank $\tau_b = 0.733$ relative to baseline. Independent-living priority retains Zurich and gives $\tau_b = 0.883$. All sixteen cities remain possible champions in each profile. At concentration 100 around baseline weights, Amsterdam's mean decision rank changes from 15.6929 to 15.6505 after its inventory supplement. The fixed education-priority recommendation remains Oslo; these are changes in the declared evidence bounds and weight distribution. The resulting differences show how the definition of a desirable city affects the decision. They do not establish measured differences between fourteen- and eighteen-year-old students.

6.3 Perturbing weights

For each preference centre w', we draw $w^{(b)} \sim \mathrm{Dirichlet}(\kappa w')$, with $\kappa \in \{20, 100, 500\}$ and 10,000 draws per setting. Every draw recomputes interval TOPSIS and maximum regret without filling missing values. We report mean decision rank, fractional first-place recommendation frequency and fractional top-three decision membership; ties share membership.

At baseline concentration 100, Oslo has a smaller sample mean decision rank (2.1092) than Zurich (2.2620), while Zurich has the greater sample recommendation frequency and is the fixed-weight minimax choice. These summaries describe different features of the chosen weight distribution. We retain minimum maximum regret as the predeclared decision rule; weight perturbations report its sensitivity. A low AHP consistency ratio checks calculation properties, not representative student opinion.

Table 4: Conditional recommendation frequencies (%) for selected cities at concentration 100.

Preference centre	Zurich	Oslo	Singapore
Baseline	43.33	27.92	28.23
Education priority	6.83	42.95	41.43
Independent living	58.99	15.46	25.12

Numerical precision can be quantified separately from evidence uncertainty. For $n = 10{,}000$ independent, identically distributed draws within each setting, Hoeffding's inequality and a union bound over the $m = 2064$ exported means across 37 settings give a common event of probability at least 0.95 with error at most $R\sqrt{\log(2m/0.05)/(2n)}$, where the response range R is 1 for frequencies and 15 for ranks [50]. The resulting half-widths are 2.38 percentage points and 0.357 ranks; shared draws across outputs are allowed. On this event, subtracting two mean intervals bounds their difference. Oslo minus Zurich at baseline concentration 100 gives $[-0.867, 0.561]$, leaving their expected mean-rank order unresolved by this guarantee. These bounds concern integration under the declared simulation laws.

6.4 Within-city fairness and growth stages

Citywide facilities per resident measure supply. Student access additionally depends on fees, opening hours, transport, disability access and location; identifying its distribution requires matched students and qualifying facilities.

A separate Shanghai school-resource case uses 16 districts, 235,542 regular senior-secondary pupils and 23,999 full-time teachers in the 2025 school year. The pooled utility is 0.604481; the student-weighted mean of district utilities is 0.603267. Ordering districts by resource utility and averaging the bottom 20% of pupil mass, with fractional inclusion of the boundary district, gives 0.580954, 0.022313 below the weighted mean. These are district-homogeneous school-location resource scenarios, not observed individual teacher exposure or resident accessibility [15]. For districts d with pupil counts n_d and pupil–teacher ratios r_d, the pupil-weighted utility is $\bar{u} = \sum_d n_d \, 15/(15 + r_d) / \sum_d n_d$. Its difference from pooled utility reflects aggregation; school-location data establish neither resident travel access nor policy effects.

For Istanbul's retained 2012 high-school cohort aged 14–18, let $L_{0.2}$ be the lowest-20% weighted mean travel utility, with fractional boundary inclusion, and μ the overall mean. Unknown utilities lie in $[0, 1]$; optional membership remains uncertain. Observed weight is $W = 499950$; including all optional members gives $N_{\max} = 681503$. The sharp joint bound is $\min(\mu - L_{0.2}) = (W/N_{\max})(\mu_{\text{obs}} - L_{0.2,\text{obs}}) = 0.226686$, where "obs" denotes observed trips. Including all optional members and setting unknown trips to 45 minutes attains it. With cohort, weights and missing-data rules fixed, allowing arbitrary utility changes to at most 5% or 20% of observed weighted mass leaves sharp minimum gaps 0.200134 or 0.129091, respectively. This fractional-mass relaxation uses declared error budgets, not estimated error rates or fractions of students. The official-portal aggregate

projection, rather than original CSV bytes, supports these historical feasibility results, not current facility-use estimates [16].

A historical Istanbul household survey offers current-high-school groups aged 14–15 and 16–18. Their travel-utility marginal bounds overlap: $[0.35961, 0.55711]$ and $[0.29347, 0.59346]$, respectively. They include unknown durations and optional members; shared unknown-age mass means their extrema need not be jointly attainable. They describe 2012 broad groups, not exact ages fourteen and eighteen today. Singapore's MOE table records 38,555 students aged fourteen and only 165 aged eighteen at the start of 2020 still in secondary schooling; the latter is not an all-eighteen-year-old resident denominator. These counts cannot be joined to census school-stage travel bins without a matching population/reference-day table [16, 17].

The shared-membership constraint permits both historical age-group orderings. For younger-minus-older mean travel utility, the joint closure is $[-0.23330, 0.26325]$, assigning the shared unknown-age mass once. Two separate completions with unknown utilities 0.05 and 0.95 give contrasts $+0.23842$ and -0.20848. Both use finite durations and allocate the shared mass to the younger group. The closure endpoints allow zero utility as a limiting value; the two reversals use strictly interior values. These are feasibility constructions within the retained 2012 aggregate uncertainty set, not observed age effects.

Comparing exact ages fourteen and eighteen requires matched resident utilities $u_{ij}^{(14)}, u_{ij}^{(18)}$ and separately elicited weights $w_j^{(14)}, w_j^{(18)}$. Our analyst preference scenarios carry no age labels.

7 Comparing cities within the same country

Every primary city belongs to a different country, so the main sample cannot identify within-country city contrasts. We use the existing United States and India cases as frozen legacy25 V117 examples with the same ten indicator definitions and scoring formula [20]. They do not enter the sixteen-city ranking, and comparison within a country does not remove differences in source year, city boundary or educational coverage.

Phoenix has higher retained affordability utilities than New York. Mumbai has the higher $PM_{2.5}$ utility, while Delhi has higher living-cost, housing, school-resource and broadband utilities in the retained evidence. The cases inherit the historical study's geographic and educational scope; their score pairs overlap, so neither case identifies a necessary overall winner. Affordability contrasts use Numbeo's 2025 named-city aggregates [21]; the Mumbai–Delhi air contrast uses 2021 WHO monitoring means [22]. These source scopes are retained in the case source index. These descriptive contrasts do not identify a national causal effect.

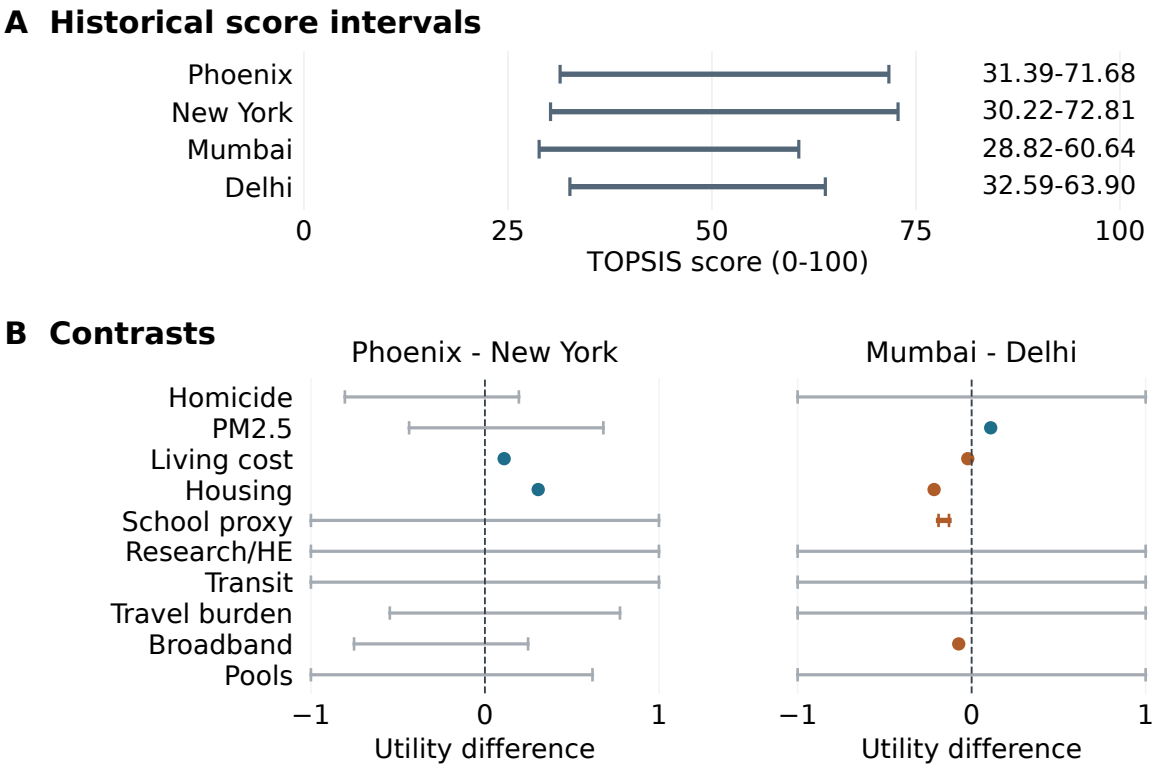


Figure 3: Frozen auxiliary V117 country cases, separate from the primary sixteen cities. Top: retained score intervals. Bottom: all twenty utility differences, first city minus second; seven exclude zero. Positive differences favour the first city; negative differences favour the second. Grey ranges leave direction unresolved. School resource retains 2024–25 school-location all-age XI–XII counts and nominal teacher-headcount allocation, not FTE. Living cost and housing use 2025 Numbeo aggregates; air uses 2021 WHO monitoring means; broadband uses the September 2026 index (July–September test window). Source years, boundaries and populations differ. Ranges are conditional identification bounds, not confidence intervals; neither pair identifies a necessary overall winner [20, 21, 22].

8 Conclusion and decision use

Zurich minimizes maximum regret under baseline and equal weights; all sixteen cities remain possible champions. Education priority selects Oslo. Withholding Zurich's homicide point selects Singapore under baseline weights and Oslo under equal weights; withdrawing its work-commute proxy selects Singapore under baseline weights but retains Zurich under equal weights. Withdrawing 28 flagged points selects Berlin and Paris, respectively.

Verified supplements narrow mean score-interval width by 14.68% while preserving supplied values. At fixed evidence, the interval revision changes 13 decision ranks and allows each city to lead in a separate completion. Families can use the recommendation as a starting point, then examine their priorities and source bounds.

A city choice built around a student's day

Under the retained team inputs, Zurich is the cautious recommendation emerging from our comparison of sixteen cities for high-school life. An alternative that places greater emphasis on education points to Oslo. Together, these findings offer students and families a practical starting point: a city comparison that connects everyday opportunities with personal priorities while keeping gaps in the evidence visible.

For a teenager, choosing a city involves the school day and the hours around it. Teaching resources, nearby higher education, clean air, safety, housing and living costs, transport, journey times, internet access and swimming facilities can all matter. Our study brings these ten considerations into one framework. Its purpose is to help families examine trade-offs, rather than assume that the same place will suit every student.

A cautious recommendation with a clear meaning

The comparison uses common scoring yardsticks and a stated balance of priorities. Where a figure is missing or describes part of a city's resources, the model keeps a range of possible scores. An unknown figure therefore remains open rather than being replaced with an invented average. Each city's overall result is a range, reflecting the information available under the model's assumptions.

The recommendation asks how far a selected city could fall behind a better-scoring alternative within those ranges. Zurich has the smallest worst-case shortfall under the baseline priorities. It also remains the cautious recommendation when all ten considerations receive equal weight. This makes it a useful place to begin a broad comparison; all sixteen cities nevertheless remain possible leaders within the retained evidence ranges. The study has not established a single certain winner.

Better evidence makes the comparison more useful

Official records become useful when their boundaries and units are clear. Vienna's broad transport register contains depot objects whose status as passenger stops is unproved. Cross-checking the operator list with a separate city map confirms 99 metro station areas. Prague's city list supplies 57 metro complexes after interchanges are counted once and a closed station is omitted. These examples show why a useful inventory needs more than a headline total.

Across twenty-three previously missing entries, additional evidence reduced the average width of the city score ranges by about 15 per cent. The original submitted values and scoring rules stayed in place. Narrower ranges make the comparison more informative, while the full set of possible leaders shows that further evidence could still change the conclusion. The records span different years, and some describe school or city resources rather than the experience of a particular teenager.

Students' priorities can change the recommendation

In an education-focused comparison, the relative emphasis on teaching resources and the density of higher-education institutions is doubled before the priorities are balanced again. Oslo then becomes the cautious recommendation. This is an illustrative choice made by the analysts, not a finding from a student survey. It gives a family a concrete way to explore how a stronger educational emphasis changes its shortlist.

City totals also need a local check. A transport-stop inventory leaves questions about walking distance, service frequency, fares and accessible routes. A pool count leaves questions about entry fees, opening hours and which neighbourhoods the facilities serve. These are useful questions for families and city planners to investigate before treating resource supply as equal access. The study does not measure an individual student's benefit from adding a stop, teacher or pool.

Use future scenarios to ask better questions

The study also examines three hypothetical changes: expanding selected resources, increasing living and travel pressures, and reducing selected educational, air-quality and travel burdens. When the affected quantities change uniformly by ten per cent, Zurich remains the cautious recommendation under both baseline and equal priorities. Repeated simulations explore what happens when the assumed changes differ across cities. They test chosen scenarios; they do not predict a future year or estimate a city's real chance of being best.

Checking a specific source can change the shortlist: treating Zurich's unverified homicide-rate figure as unknown points to Singapore under the original priorities and Oslo with equal priorities. Separately setting aside Zurich's general work-commute proxy also points to Singapore under the original priorities; equal priorities retain Zurich. The supplied figures remain in the baseline; these checks test which information the recommendation depends on.

A family's next step is to pair the city comparison with its own school options, budget, daily route and preferred activities. Students can give education more emphasis, examine the affordability trade-offs, and check whether nearby opportunities fit their routines. The model makes those choices easier to discuss: a clear starting recommendation, room for different priorities, and an explicit view of the evidence needed for a more personal decision.

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Report on Use of AI Tools

Documentation status: incomplete review record. This appendix records assistance supported by saved task metadata and research receipts. The complete team AI history remains to be reconciled with actual saved conversations and other tool records before submission.

Known tool and scope

OpenAI Codex, used 8–10 November 2026. The task log records Codex Desktop client version `0.166.0-alpha.5.1` and model identifier `gpt-6.1-sol` in all 74 saved root turn contexts within the inspected prefix. The identifier does not expose an immutable serving build or establish the model used by every concurrent assistant. The assistant supported official-source search and extraction, data-scope checks, Python calculations and audits, LaTeX drafting, and wording revisions. Concurrent assistant tasks prepared bounded evidence checks, a controlled model diagnostic and the news-article contribution [18]. The anti-defensive-writing skill supplied editorial instructions; it is a workflow resource, not a surveyed student preference or a data source.

Exa, used 8–9 November 2026 through the recorded `web_fetch_exa` tool, retrieved and extracted text from official webpages, PDFs and data interfaces [19]. Saved receipts identify the service, source URLs and returned text, but do not specify an internal model or service version. They do not establish whether a generative model participated in every extraction. Extracted representations are distinguished from original downloaded bytes. For example, Tokyo sports-facility PDF text with ambiguous row order was excluded; the admitted five-venue lower bound uses separately downloaded Nerima municipal HTML.

Recovered records and coverage

The supporting inventory authenticates a complete-line prefix of this root task log by byte count and SHA-256. It ends at 14:31:18.850 UTC on 9 November 2026 (22:31 in Singapore); subsequent work is outside the window. A private export retains 326 selected visible text records: six selected genuine requests, 259 progress messages and 61 complete final responses. The original 125-record selection and its private byte prefix are preserved.

A second private export retains 1,865 plaintext assistant tool-call inputs: 1,749 custom-call input strings and 116 function-call argument strings. These preserve generated code and LaTeX where present in the calls, extending the evidence beyond visible messages. Recorded inputs do not establish that each invocation succeeded or that every nested tool ran.

This capture covers one root session. It excludes tool results, attachment contents, sub-agent response sessions, other chats and other team members. Reasoning, system and developer messages, encrypted fields, internal metadata and injected continuations are excluded. Public inventories contain metadata and record hashes; the original log and exported text remain private and are not included as submission transcripts. No missing prompt or output has been reconstructed and labelled as an exact historical quotation.

Verification associated with this draft

The research calculations preserve the original city CSV and the V181 primary evidence. The V181 validation independently recomputes weighted norms and self-excluded competitor maxima, verifies interval membership of constructed completions and checks source hashes. Original provider records are kept separately from derived tables. Compilation and rendered-page review check the presentation. These checks establish the specified calculation and document properties; they do not replace the team's responsibility to review factual claims, original supplied data and final authorship.

Records required to finalize this report

The final report must identify every AI tool actually used, its available model or version, date and purpose, and associate relevant use with the text and references. Actual saved query and response records should be inserted according to the applicable COMAP policy, including its stated exceptions. Other team members' tools, translation use and earlier conversations remain to be confirmed from their records.

The current in-text disclosure and references identify Codex and Exa assistance. Submission readiness also requires a verified Team Control Number in place of the review header and confirmation that this report accounts for the team's actual use.